



Cite this: DOI: 10.1039/c5cc04753f

Received 9th June 2015,
Accepted 14th August 2015

DOI: 10.1039/c5cc04753f

www.rsc.org/chemcomm

A crystalline low molecular weight cyclic organoboron compound for efficient solid state lithium ion transport†

Prerna Joshi,^{ab} Raman Vedarajan^a and Noriyoshi Matsumi^{*a}

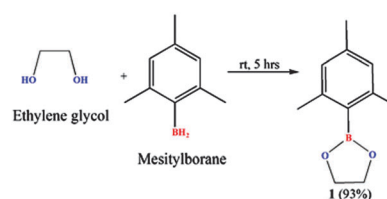
We report the synthesis of a crystalline cyclic organoboron compound that shows an anomalous Li-ion conduction behaviour with specific composition and method of insertion of a Li salt. The enhanced ionic conductivity of crystalline boric ester/LiTFSI prepared by the grinding method should be due to the formation of regulated ion-conduction channels scaffolded by boron–anion interaction.

Organic liquid electrolytes have long been used in LiBs due to their impressive ionic conductivity and practical usage at ambient temperature.¹ However, non-flammability and a simple insertion procedure of the electrolyte during the manufacture of LiBs make solid polymer electrolytes (SPEs) attractive alternatives for use in LiBs.² Nevertheless, there were demerits associated with SPEs, ranging from poor ionic conductivity, low lithium ion transference number, low mechanical strength *etc.*³ In this context, various types of SPEs broadly classified as gel-type polymers,^{4,5} solvent free polymers,^{5,6} inorganic crystalline compounds and inorganic glasses have mushroomed in the research arena.⁵ In general, higher ionic conductivity and the ease of processability, when compared to crystalline electrolytes, allow extensive use of organic polymer type electrolytes in LiBs. In these organic polymer electrolytes, the segmental motion of polymer chains is considered to be responsible for the ion transport mechanism. However, the theoretical limit² of ionic conductivity based on the segmental motion has been predicted to be 10^{-3} to 10^{-4} S cm⁻¹. This value of ionic conductivity is a couple of orders lower compared to its liquid counterpart.⁵

In 2011, inorganic crystalline electrolytes such as lithium ion super conductors were projected as major competitors to the organic electrolytes with an ionic conductivity of 10^{-3} S cm⁻¹ at 50–80 °C.³ As another unique approach,⁷ Leclerc *et al.* reported that anion doped π -conjugated systems without any radical carrier

species exhibited high conductivity. However, electronic conduction was not prohibited in their systems and it was believed that lithium ions were attached to the main chains, and their high mobility could be due to an interaction with the delocalized anion on π -conjugated polymers.^{8,9} Recently metal organic frameworks have also come into the frame, though mostly as electrode materials. However, the use of metal organic frameworks as electrolyte materials has been reported by Wiers *et al.*, who said that lithium alkoxide based MOFs can lead to high ionic conductivity at ambient temperatures as open metal centres can bind to the anions leaving the Li⁺ ions free to move along the channels.¹⁰ In order to override the limitation caused due to segmental motion in ordinary SPEs and to exceed the theoretical limit, there is a need to seek an altogether new ion conductive mechanism that is different from ordinary polymer electrolytes. Hence, a need for a low molecular weight solid electrolyte resulting in better electrochemical properties compared to aforementioned electrolytes led us to design a new crystalline cyclic organoboron compound for lithium ion conduction. In the present work, for the first time a low molecular weight cyclic organoboron compound was investigated with respect to its ion conductive properties. This paper details the synthesis of the compound with a change in the mode of insertion of a Li salt into the crystalline electrolyte, which resulted in drastic changes in ion conductive behaviours.

An equimolar amount of dehydrated ethylene glycol (0.598 mL, 0.0107 moles) was added to the synthesized mesitylborane¹¹ (1.42 g, 0.0107 moles) at 0 °C and the reaction mixture was kept stirring at room temperature for 5 hours under a N₂ atmosphere (Scheme 1). The resulting mixture was then washed with hexane and dried under vacuum for 3 hours to obtain a crystalline white powder. The final



Scheme 1

^a School of Materials Science, Japan Advanced Institute of Science and Technology, Ishikawa, Japan. E-mail: matsumi@jaist.ac.jp

^b M.Tech (Chemical Synthesis and Process Technologies), Department of Chemistry, University of Delhi, North Campus, Delhi, India

† Electronic supplementary information (ESI) available. See DOI: 10.1039/c5cc04753f



Fig. 1 Crystals of **1** (EGMB).

product, denoted as **1** (EGMB), was recrystallized using hexane to obtain needle shaped crystals (Fig. 1).

The synthesized low molecular weight cyclic organoboron electrolyte, **1** (EGMB), was characterized by ^1H and ^{11}B -NMR in $\text{DMSO}-d_6$ at 400 MHz. A single peak at 31.3 ppm referring to the incorporation of boron as a boric ester in ^{11}B -NMR confirmed the presence of a single pure environment of boron. ^1H -NMR confirmed the structure of **1** (EGMB).

[^1H -NMR (400 MHz, $\text{DMSO}-d_6$): 6.79–6.72 (phenyl ring protons), 4.30 ($-\text{CH}_2-$ groups of ethylene glycol), 3.33 (H_2O), 2.50 ($\text{DMSO}-d_6$), 2.27–2.08 ($-\text{CH}_3$ protons of the mesityl group)]. A cyclic organoboron electrolyte was found to be thermally stable up to 102 $^\circ\text{C}$ and no

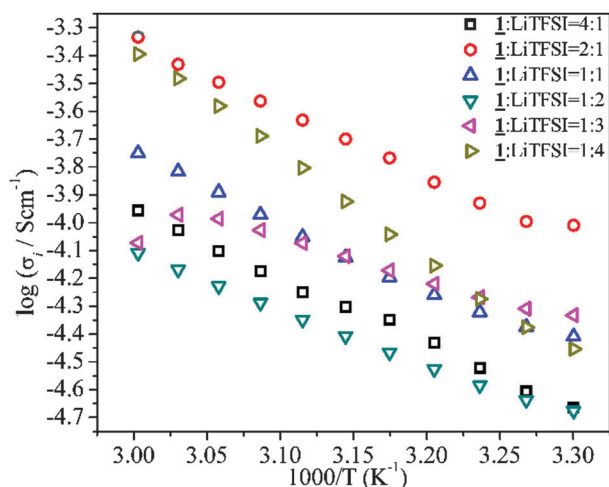


Fig. 2 Arrhenius plots for samples doped with LiTFSI by the conventional method of dissolving in THF.

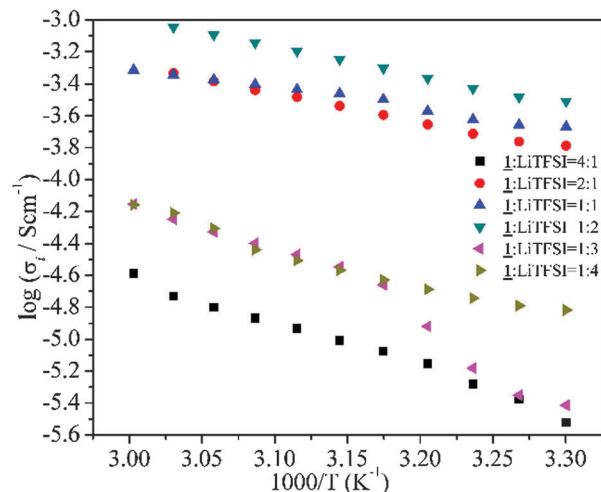


Fig. 3 Arrhenius plots for samples doped with LiTFSI by the grinding method.

change in thermal properties was observed after the addition of a lithium salt, lithium bis(trifluoromethanesulfonyl)imide (LiTFSI). The Raman spectrum (shown in the ESI †) of **1** was also studied which was similar to the simulated Raman spectrum obtained using Gaussian 09 software. HRTEM images (provided in the ESI †) of the needle crystals of **1** showed lattice fringes which can be associated with an organic crystal.

For further electrochemical analysis, **1** (EGMB) was doped with LiTFSI. The physical aspect of **1** (EGMB) changed from a transparent crystal to an opaque powder. Addition of the Li salt was carried out by two different methods. In the first case, the conventional method of inserting the salt using a solvent (THF) was employed. In the second case, a rather simple grinding of **1** (EGMB) with LiTFSI was adapted. Fig. 2 and 3 shows the Arrhenius plots of samples prepared by the conventional method and the grinding method, respectively. Most samples prepared by conventional methods showed a monotonous increase in ionic conductivity

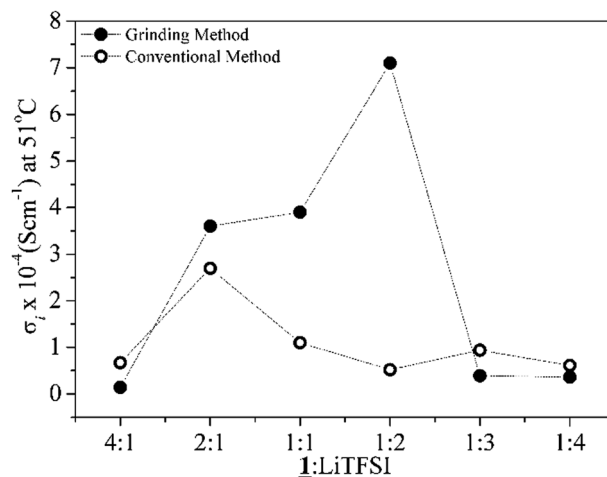


Fig. 4 Comparison between ionic conductivities of samples prepared by grinding and conventional methods by dissolving in solvent at various lithium salt concentrations at 51 $^\circ\text{C}$.

Table 1 Comparison of VFT parameters for the samples prepared by the conventional method and grinding

		Grinding method					Conventional method				
S. no.	Sample (1:LiTFSI)	B				σ_i at 51 °C (S cm ⁻¹)	B				σ_i at 51 °C (S cm ⁻¹)
		A (S cm ⁻¹ K ^{1/2})	(K)	(eV)	R ²		A (S cm ⁻¹ K ^{1/2})	(K)	(eV)	R ²	
1	4:1	2.52	1607	0.1385	0.9997	1.4×10^{-5}	7	4.74	1555	0.1340	6.6×10^{-5}
2	2:1	6.90	1209	0.1042	0.9992	3.6×10^{-4}	8	4.81	1583	0.1361	2.7×10^{-4}
3	1:1	1.72	765.3	0.0659	0.9995	3.9×10^{-4}	9	59.2	1796	0.1548	1.1×10^{-4}
4	1:2	15.7	1236	0.1065	0.9994	7.1×10^{-4}	10	2.17	1350	0.1163	5.2×10^{-5}
5	1:3	33.7	1707	0.1471	0.9998	3.9×10^{-5}	11	1.25	1071	0.0922	9.4×10^{-5}
6	1:4	0.66	1331	0.1147	0.9992	3.6×10^{-5}	12	1.32	1232	0.1062	6.1×10^{-5}

with increase in temperature at a relatively low concentration of lithium salt (Arrhenius behaviour). On the other hand, the samples prepared by the grinding method showed different behaviours with lower activation energy of ion transport. This implies that different ion conduction mechanisms were operating. Another feature observed in these samples was that the ionic conductivity initially increased with increase in the Li salt concentration and decreased drastically after an optimum concentration of lithium salt. In the first case, the maximum ionic conductivity was observed to be 2.7×10^{-4} S cm⁻¹ at 51 °C while the maximum ionic conductivity observed by the facile grinding method reached 7.1×10^{-4} S cm⁻¹ (51 °C).

The conductivities shown by the samples prepared by the grinding method were significantly higher at the optimal concentration of lithium salt which can be due to the retention of crystallinity of a Lewis acidic ordered structure by grinding which otherwise might be lost in the case of conventional samples due to solvation (Fig. 4 and Table 1).

The lower values of ionic conductivity at lower concentration of lithium salt can be attributed to the minimal availability of ions for conduction. The lowering of ionic conductivity at higher concentration of salt can be due to an increased activation energy required for ion dissociation. The high conductivity seen at an optimal concentration of lithium salt in the sample prepared by grinding can be hypothesized to be due to the formation of nanochannels^{12–15} by an arrangement of anions scaffolded by the undisturbed crystalline nature of the sample through boron anion interaction,^{16–19} allowing easier and faster transportation of ions by the ion hopping mechanism. This becomes evident from the fact that ionic conductivity for a composition of 1:LiTFSI = 1:2 prepared by the conventional method is very low as compared to the sample with the same composition prepared by grinding. This can be due to the absence of any such nanochannel formation in the sample prepared by conventional method.

A closer look at the VFT parameters of 3 samples prepared by the grinding method (samples 2, 3 and 4) showed very low activation energy of ion transport (B). This led to markedly higher ionic conductivity. On the other hand, the samples prepared by the conventional method showed relatively high activation energy and low ionic conductivity at the same composition. As a specific instance, comparison between grinding and conventional methods for a composition of 1:LiTFSI = 1:1 shows that although the carrier ion number in the sample prepared by grinding is only 1.72, its ionic conductivity is 3.5 times higher than that of the

sample prepared by the conventional method with the same composition exhibiting a high carrier ion number, 59.2. Hence, in the case of grinded samples, low activation energy of ion transport resulted in high ionic conductivity. This evinces our hypotheses of formation of special channelized pathways resulting in low activation energy allowing facile transportation of Li⁺ ions. Some samples prepared by the conventional method (samples 10, 11 and 12) also show low activation energy of ion transport (B), at high concentration of LiTFSI. However, enhancement of ionic conductivity was not observed for these systems due to a small carrier ion number (A).

The cell employed for measuring the lithium ion transference number²⁰ was a symmetrical cell consisting of a Li/electrolyte/Li configuration. The experimental procedure and the calculations are detailed in the ESI.† The lithium transference number was determined to be 0.28. The potential window of the sample measured by cyclic voltammetric analysis in a 3E configuration cell (using the Pt–Pt electrode (wire) vs. Ag/AgNO₃) was found to be 5.3 V.

A low molecular weight cyclic organoboron crystalline compound was synthesized and characterized for its structure and other thermal properties. The electrochemical analysis of the sample showed a high ionic conductivity of 7.1×10^{-4} S cm⁻¹ at an optimized concentration of lithium salt. A facile grinding technique of the lithium salt addition with the prepared organoboron compound resulted in high ionic conductivity compared to the conventional method of mixing by dissolution and evaporation with solvents. The presence of crystallinity in Lewis acidic EGMB can be considered as an important factor behind the high ionic conductivity of the electrolyte prepared by grinding due to a more regulated ion conduction path in the matrices.

Notes and references

- 1 A. Andersson, Surface Phenomena in Li-Ion Batteries, Acta Universitatis Upsaliensis, *Comprehensive Summaries of Uppsala Dissertations from the Faculty of Science and Technology*, Uppsala, 2001, p. 60.
- 2 M. Wakiyama, Y. Kadoma, N. Kumagai, H. Mita, R. Araki, K. Ozawa and Y. Ozawa, *J. Solid State Electrochem.*, 2012, **16**, 847–855.
- 3 N. Kamaya, K. Homma, Y. Yamakawa, M. Hirayama, R. Kanno, M. Yonemura, T. Kamiyama, Y. Kato, S. Hama, K. Kawamoto and A. Mitsui, *Nat. Mater.*, 2011, **10**, 682–686.
- 4 A. Manuel Stephan, *Eur. Polym. J.*, 2006, **42**, 21–42.
- 5 M. Park, X. Zhang, M. Chung, G. B. Less and A. M. Sastry, *J. Power Sources*, 2010, **195**, 7904–7929.
- 6 M. A. Ratner and D. F. Shriver, *Chem. Rev.*, 1988, **88**, 109–124.
- 7 T. Yong, J. Wang, Y. Mai, X. Zhao, H. Luo and L. Zhang, *J. Power Sources*, 2014, **254**, 29–32.

- 8 N. Matsumi, M. Nakashiba and H. Ohno, *Polym. Bull.*, 2003, **50**, 259–264.
- 9 M. Ranger and M. Leclerc, *Macromolecules*, 1999, **32**, 3306–3313.
- 10 B. M. Wiers, M.-L. Foo, N. P. Balsara and J. R. Long, *J. Am. Chem. Soc.*, 2011, **133**, 14522–14525.
- 11 K. Smith, A. Pelter and Z. Jin, *Angew. Chem., Int. Ed. Engl.*, 1994, **33**, 851–853.
- 12 S. Vorrey and D. Teeters, *Electrochim. Acta*, 2003, **48**, 2137–2141.
- 13 Z. Gadjourova, Y. G. Andreev, D. P. Tunstall and P. G. Bruce, *Nature*, 2001, **412**, 520–523.
- 14 T. Nakamura, T. Akutagawa, K. Honda, A. E. Underhill, A. T. Coomber and R. H. Friend, *Nature*, 1998, **394**, 159–162.
- 15 S. B. R. S. Adnan, *Int. J. Electrochem. Sci.*, 2012, **7**, 9844–9858.
- 16 N. Matsumi, M. Nakashiba, T. Mizumo and H. Ohno, *Macromolecules*, 2005, **38**, 2040–2042.
- 17 N. Matsumi, K. Sugai and H. Ohno, *Macromolecules*, 2002, **35**, 5731–5733.
- 18 N. Matsumi, K. Sugai, M. Miyake and H. Ohno, *Macromolecules*, 2006, **39**, 6924–6927.
- 19 T. Mizumo, K. Sakamoto, N. Matsumi and H. Ohno, *Electrochim. Acta*, 2005, **50**, 3928–3933.
- 20 J. Evans, C. A. Vincent and P. G. Bruce, *Polymer*, 1987, **28**, 2324–2328.